

Pranav Kunnamkudath Roy

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EDUCATION

SRM Institute of Science and Technology

Kattankulathur, Tamil Nadu

B.Tech, CSE – Computer Networking

2023 – 2027

- CGPA: 8.37/10

Mount St. Mary's School

New Delhi

Class XII, CBSE

2023

- Percentage: 75.4%

Mount St. Mary's School

New Delhi

Class X, CBSE

2021

- Percentage: 91.4%

PROJECTS

RWM – Rust Window Manager | *Rust, x11rb, X11* | github.com/ReitardXd/rwm

2025

- Built a tiling window manager from scratch in Rust using x11rb raw X11 bindings, implementing the WM protocol end-to-end: SUBSTRUCTURE_REDIRECT takeover, ICCCM graceful close (WM_DELETE_WINDOW with XKillClient fallback), and EWMH title tracking.
- Designed a single-threaded, non-blocking event loop (200ms idle poll, 1Hz status-bar clock) with master-stack tiling across 9 workspaces; fixed a workspace-switching race via a pending-unmap queue – 1,057 LOC, 1 crate dependency, 0 unsafe blocks.

Seca – Social Reading Platform | *React, Node.js, Express, MongoDB, Groq API* | github.com/ReitardXd/seca | Live Demo2024

- Built a full-stack social reading platform (React 19/Vite/Tailwind client, Express 5/Mongoose API, MongoDB Atlas) with Google OAuth 2.0 and JWT authentication across 5 database models; deployed client on Vercel and server on Render.
- Integrated the Open Library API (with MongoDB caching) and Groq (Llama 3) for AI-generated quizzes with resilient JSON-parse recovery; implemented invite-code groups, reading streaks, and a per-group leaderboard – 2,136 LOC across 58 files.

N-Body Gravitational Simulator | *C++, OpenGL, GLSL, GLFW, OpenMP* | github.com/ReitardXd/nbody

2024

- Built a real-time 3D N-body gravitational simulator in C++17/OpenGL 3.3 modeling colliding spiral galaxies and a black hole, implementing a Barnes-Hut octree with center-of-mass multipole approximation ($\theta = 0.5$) to reduce force computation from $O(N^2)$ to $O(N \log N)$.
- Parallelized octree traversal with OpenMP (lock-free, read-only tree), implemented a symplectic velocity-Verlet integrator with Plummer softening, and built a custom OpenGL renderer with HDR bloom, point-sprite particles, and fading trails.

NERVE – Network Incident Response Pipeline | *Python, Cisco DNAC API, Groq LLaMA, Streamlit, SQLite*2025

- Built an autonomous network incident-response pipeline integrating Cisco DNA Center REST APIs, Groq LLaMA for root-cause analysis, Discord webhook alerting, and SQLite-backed ticketing with automated MTTR tracking.
- Designed an edge-triggered state machine in the polling loop that opens tickets only on healthy-to-unhealthy transitions, deduplicates per device, and auto-resolves on recovery; managed DNAC token lifecycle with hourly refresh and forced re-auth on failure.

TECHNICAL SKILLS

Languages: C++, C, Rust, Python, JavaScript, Bash/Shell, SQL

Systems & Networking: POSIX, Linux (Arch), TCP/IP, X11, Cisco DNAC API

Frameworks & Tools: FastAPI, Streamlit, OpenGL/GLFW, Node.js, PostgreSQL, Docker, Git, Neovim, GDB

Security: HackTheBox, CTF, hashcat, vulnerability research

Competitive Programming: Codeforces – Expert 1654 rating

PUBLICATIONS

Smart Sleep Posture Monitoring System Using Embedded Sensors and IoT

May 2025

- International Research Journal on Advanced Engineering and Management
- ESP32-based sleep posture classifier using MPU6050 IMU and FSR sensors with ML classification and vibration feedback.

LANGUAGES

English (Native) | Hindi (Native) | Tamil (Professional Working) | Malayalam (Professional Working)